WORKED SOLUTION

STEP I 2006, Question 10

Mechanics • Projectile Motion

STEP I 2006, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q10

The key idea

The trajectory is a parabola and the roof is a straight line, so "just touches" means the line is a *tangent*: substituting one into the other gives a quadratic in x with a repeated root. The sign question is settled by asking where that repeated root lies. For the final part, the particles can only meet where P meets the roof, so Q must arrive at that point at exactly the moment P does; comparing horizontal positions at that time gives the result.

Solution

The trajectory

Model P as a particle moving freely under gravity, with no air resistance. Its initial velocity components are $V\cos 45^\circ = V/\sqrt{2}$ both horizontally and vertically, so at time t

$$x = \frac{Vt}{\sqrt{2}}, \qquad y = \frac{Vt}{\sqrt{2}} - \tfrac{1}{2}gt^2.$$

Eliminating $t = \sqrt{2}\,x/V$ gives $\frac{1}{2}gt^2 = gx^2/V^2$, so the trajectory is

$$y = x - \frac{gx^2}{V^2}.$$

Touching the roof

P meets the roof where $x - gx^2/V^2 = x\tan\alpha + b$, that is, where

$$\frac{g}{V^2}\,x^2 + (\tan\alpha - 1)\,x + b = 0. \qquad (*)$$

Touching (rather than crossing) means $(*)$ has a repeated root, so its discriminant is zero:

$$(\tan\alpha - 1)^2 = \frac{4gb}{V^2} \quad\Longrightarrow\quad V(\tan\alpha - 1) = \pm 2\sqrt{bg}.$$

Ask yourself Both signs satisfy the discriminant condition. What tells them apart?

The repeated root of $(*)$ is

$$x_0 = -\frac{(\tan\alpha - 1)V^2}{2g} = \frac{(1 - \tan\alpha)V^2}{2g}.$$

The particle moves in the direction of increasing x (since $x = Vt/\sqrt{2}$), so it can only touch the roof where $x_0 > 0$. That forces $\tan\alpha - 1 < 0$, and as $V > 0$ the left-hand side $V(\tan\alpha - 1)$ is negative. Hence

$$V(-1 + \tan\alpha) = -2\sqrt{bg},$$

as required. (With the positive root, the tangent point would be at $x_0 < 0$, on the backward continuation of the parabola, which P never visits.) Geometrically, the roof must be less steep than the initial direction of motion, $\alpha < 45°$; otherwise the roof stays above the line $y = x$, and the parabola lies below that line for $x > 0$.

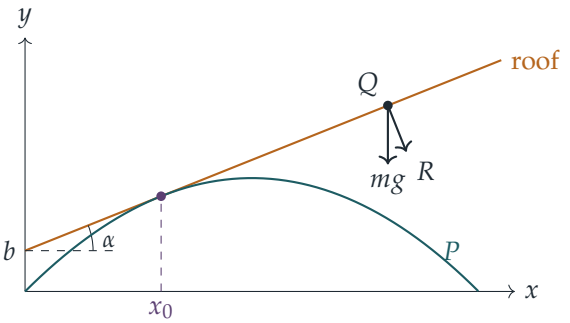
The time of touching

Since $x = Vt/\sqrt{2}$, the touching time is

$$T = \frac{\sqrt{2}\,x_0}{V} = \frac{\sqrt{2}\,(1 - \tan\alpha)V}{2g} = \frac{V(1 - \tan\alpha)}{\sqrt{2}\,g}.$$

The particle on the rail

Model Q as a particle on a smooth rail along the straight line $y = x\tan\alpha + b$; it is projected from $(0, b)$ along the rail in the direction of increasing x (it has to be, to reach P, whose x is always positive). The forces on Q are its weight mg and the normal reaction R of the rail, perpendicular to the rail. The figure shows these forces, together with P's path touching the roof at $x = x_0$.



There is no friction, so resolving along the rail, up the slope,

$$m\ddot{s} = -mg\sin\alpha,$$

where s is the distance of Q along the rail from $(0, b)$. With $s = 0$ and $\dot{s} = U$ at $t = 0$,

$$s = Ut - \tfrac{1}{2}gt^2 \sin\alpha, \qquad x_Q = s\cos\alpha.$$

P is never on the roof except at the touching point, and Q is always on the roof, so the particles can touch only if Q is at that point at time T. Both particles are then on the roof line, so it is enough that their x-coordinates agree at time T:

$$\left(UT - \tfrac{1}{2}gT^2 \sin\alpha\right)\cos\alpha = \frac{VT}{\sqrt{2}}.$$

Dividing by $T > 0$ and substituting $gT = V(1 - \tan\alpha)/\sqrt{2}$:

$$U\cos\alpha = \frac{V}{\sqrt{2}} + \tfrac{1}{2}\sin\alpha\cos\alpha \cdot \frac{V(1-\tan\alpha)}{\sqrt{2}}.$$

Multiplying by $2\sqrt{2}$, and using $\sin\alpha\cos\alpha\tan\alpha = \sin^2\alpha$,

$$\begin{aligned} 2\sqrt{2}\,U\cos\alpha &= 2V + V\sin\alpha\cos\alpha\,(1-\tan\alpha) \\ &= V\left(2 + \sin\alpha\cos\alpha - \sin^2\alpha\right), \end{aligned}$$

which is the required result.

Checks

- Units: $V(1-\tan\alpha)/g$ is a speed divided by an acceleration, so a time; and $\sqrt{bg}$ is a speed, as is $V(\tan\alpha - 1)$.
- Limiting case $\alpha \to 0$ (a flat roof): $b = V^2/(4g)$ is the familiar maximum height $(V\sin 45^\circ)^2/(2g)$, reached at $T = V/(\sqrt{2}g)$; and Q feels no force along the rail, so it must match P's horizontal speed, $U = V/\sqrt{2}$. The formula gives exactly this.
- Numerically, with $g = 9.8$, $V = 10$ and $\alpha = 0.3$ rad, the touching point is at $x_0 \approx 3.524$. Stepping Q along the rail with the derived U for random V and $\alpha < \pi/4$ puts it at P's position at time T to within 10^{-6}, and P stays below the roof at all other times.

Common slips

- Giving the acceleration of Q as $-g$ or $-g\cos\alpha$: only the component of gravity along the rail, $g\sin\alpha$, acts along it.
- Treating U as horizontal speed: U is the speed along the rail, so Q's horizontal displacement is $s\cos\alpha$.
- Justifying the negative root by saying "speeds are positive" only. It is the sign of x_0 (equivalently $\tan\alpha < 1$) that decides it.

Beyond STEP

The tangency condition is a special case of a general idea: the boundary of the region a family of curves can reach is their *envelope*, and finding it amounts to asking when a quadratic has a repeated root. For projectiles launched at a fixed speed in all directions, the envelope is the "parabola of safety". The motion of Q is the simplest constrained motion; first-year courses handle constraints systematically with Lagrangian mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers both projectiles and constrained motion carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.